

JIKAI WANG

✉ wangjikai2025@gmail.com · 🎓 Google Scholar · 🌐 wang-jikai

RESEARCH INTERESTS

◇ Climate Risk Propagation
◇ Systemic Resilience

◇ Coupled Human–Environment Systems
◇ Climate Transition Policy

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Aug. 2023 – Feb. 2026

Master of Engineering in Green Growth and Sustainability

GPA: 3.95/4.30 (96.5/100)

Advisor: Prof. Jiyong Eom

Relevant Coursework: Energy and Resource Economics (A^+); Climate Policy and Finance (A); Project for Carbon Neutrality Assessments (A); Science of Climate Change (A)

Southwest Jiaotong University (SWJTU), Chengdu, China

Sep. 2019 – Jun. 2023

Bachelor of Science in Statistics

GPA: 86.21/100

Research Advisor: Prof. Gaoxiu Qiao

Relevant Coursework: Econometrics (A); Time Series Analysis (A^+); Multivariate Statistics (A^+); Stochastic Processes (A^-); Real Analysis (A^-)

PEER-REVIEWED PUBLICATIONS

1. **Jikai Wang**; Jiyong Eom. Wind power forecast uncertainty co-varies across Korean regions through asymmetric heavy-tailed connectedness networks *Communications Earth & Environment*, 2026
Oral Presentation at Korean Operations Research and Management Science Society Fall Conference, Nov 2025
2. **Jikai Wang**; Gaoxiu Qiao. Extreme events and quantile time-frequency volatility connectedness across crude oil, green bonds and low-carbon equity markets. *Research in International Business and Finance*, 2025.
3. **Jikai Wang**; Kai Feng; Gaoxiu Qiao. A Hybrid Deep Learning Model for Bitcoin Price Prediction: Data Decomposition and Feature Selection. *Applied Economics*, 2024.
4. Yang Hu, Yanran Hong, Kai Feng, and **Jikai Wang**. Evaluating the Importance of Monetary Policy Uncertainty: The Long- and Short-Term Effects and Responses. *Evaluation Review*, 2023.

RESEARCH EXPERIENCE

Transmission Failure of Unitary-Planner Carbon Pricing in a Heterogeneous Financial Economy and Policy Instrument Remediation

Independent Researcher

Dec. 2025 - Present

Objective: Investigating why the optimal carbon price produced by DICE fails to translate into technology transition in a decentralized economy, and identify transmission bottlenecks and potential remedial policy tools.

- Built a bidirectional coupling between DICE2023 and the macrofinancial agent-based model FinDK via backstop technology cost parameters to transmit richer carbon price path information out of mutual consistency perspective.
- Constructing a firm-level clean transition funnel with counterfactual experiments to locate bottlenecks across four candidate mechanisms (information transmission, cash flow, bank credit rationing, and technology menu), and designing a policy portfolio spanning production-side flow taxes to acquisition-side state-dependent feebates to improve decarbonization intensity.

Climate-Driven Shock Propagation in Regional Energy Networks

Research Assistant, KAIST, Advisor: Prof. Jiyong Eom

Nov. 2024 – Dec. 2025

Objective: Investigated how meteorological disturbances manifest as localized multi-dimensional forecast uncertainty and become systemic through cross-regional co-dependence and sequential network activation.

- Characterized regional forecast-error distributions using volatility, skewness, and kurtosis to distinguish dispersion, asymmetry, and tail heaviness across main wind-power regions.
- Constructed dimension-specific networks using R^2 connectedness, revealing that systemic risk structure varied by uncertainty structure rather than aggregate intensity alone.

- Applied a Ripple Spreading Network Model to simulate sequential risk activation, translating static cross-regional dependence into dimension-specific propagation paths and dynamic node-priority rankings.

Financial Network Resilience under Socioeconomic and Environmental Shocks

Research Intern, SWJTU, Advisor: Prof. Gaoxiu Qiao *June 2023 – May 2025*

Objective: Investigated whether shock-induced cross-market volatility linkages reflect temporary synchronization or persist as structural sources of systemic vulnerability.

- Constructed mean- and quantile-based time-frequency connectedness networks using TVP-VAR and QVAR to trace linkage dynamics across investment horizons and market states.
- Introduced new indicators, the Connectedness Survival Rate (CSR) and Quantile CSR (QCSR), to quantify linkage persistence across adjacent periods, extending connectedness analysis from magnitude to temporal persistence.
- Converted CSR/QCSR into portfolio-allocation signals; in backtests, both strategies outperformed single-asset benchmarks on risk-return metrics, while QCSR achieved returns comparable to CSR with a lower maximum drawdown.

Topological Formation and Evolution of Higher-Order Interactions in Social Networks

Research Assistant, National Taiwan University & Lehigh University
Advisor: Prof. Hsuan-Wei Lee *Apr. 2023 – Mar. 2024*

Objective: Investigated how three-person group interactions form and evolve over time, and identified factors that drive their emergence.

- Built temporal networks from high school Facebook data, modeling joint interactions among three users as group-level relations.
- Applied persistent homology to track the emergence, persistence, and dissolution of higher-order structures, identifying shifts around semester starts and holidays.
- Found that low-effort actions such as likes typically preceded higher-effort interactions such as tags, suggesting that cohesive group interactions develop gradually.

PROFESSIONAL EXPERIENCE

ENCORED Inc. Seoul, South Korea *Feb 2026 - Present*
Researcher, Electricity Trading Team, focusing on virtual power plant (VPP) and wind & solar power policy

SELECTED HONORS & AWARDS

KAIST Full Scholarship	<i>2023–2026</i>
Top 100 Undergraduate Thesis Award , Southwest Jiaotong University	<i>2023</i>
One of 100 undergraduate theses selected university-wide.	
Outstanding Graduate Award , Southwest Jiaotong University	<i>2023</i>
Emei 89 Carrying Forward China Scholarship , Southwest Jiaotong University	<i>2022</i>
One of 20 university-wide recipients selected for academic excellence.	

SERVICES

Teaching Assistant , Managerial Economics of Climate & Energy Technologies	<i>Spring 2026</i>
Teaching Assistant , Scientific Writing	<i>Spring, Fall 2025</i>
Teaching Assistant , Climate Technopreneurship	<i>Fall 2024</i>
Administrative Assistant , Department of Green Growth and Sustainability	<i>Spring, Winter 2024</i>

SKILLS

Methods	Network Analysis, Integrated Assessment Modeling, Agent-Based Modeling, Econometrics
Programming	Python, R, Julia, MATLAB
Languages	English (IELTS 7.5); Chinese (Native); Korean (Basic)